

Gap Analysis: Isaac AI Capabilities vs. GC Influencer Signals

Summary

Isaac has **8 capability types**. Cross-referencing against signals from TBS, PSPC, SSC, and HC/PHAC EA identifies **7 material gaps** and **3 re-framing needs**.

Isaac's 8 Capabilities (Current State)

1. Generative AI (Summarization, drafting, copilots)
 2. Knowledge AI (RAG assistants, semantic search)
 3. Predictive AI (Forecasting, anomaly detection)
 4. Classification AI (Automated tagging, triage)
 5. Conversational AI (Assistants, chatbots)
 6. Perception AI (OCR, vision, speech processing)
 7. Orchestration AI (Agentic workflows, task automation)
 8. Decision Support AI (Recommendations, prioritization)
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TIER 1: CRITICAL GAPS — Signaled by Multiple Stakeholders

Gap 1.1: AI Model Management & Registry

Signaled by:

- **PSPC**: "Capability-based vendor categorization" requires vendors to declare model versions, update cadence, rollback procedures
- **HC/PHAC EA**: "Platform reuse & enterprise AI readiness" requires model versioning, governance, lineage tracking
- **ISO 42001**: AI management systems require model inventory, control of AI changes

(§5.4)

Why missing: Isaac focuses on AI *use* (what it does), not AI *lifecycle* (how it's managed).

What's needed:

- Sub-capability: **Model Registry & Version Control**
 - Scope: Model provenance, training data lineage, performance benchmarks, rollback plans
 - Governance: Change control per ISO 42001 §5.4.7
 - Example: "Model XYZ v2.3 (trained on Dec 2025 data, accuracy 94.2%, rollback to v2.1 available)"

Artifact:

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AI Model Management Capability
├─ Model Registry (versioning, lineage, metadata)
├─ Training Data Governance (provenance, update control)
├─ Model Change Control (approval, testing, rollback)
└─ Performance Monitoring (accuracy decay, drift detection)
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Gap 1.2: AI Explainability & Interpretability

Signaled by:

- **TBS ADM Directive** (§3.2): "Explanation of an automated decision" is *mandatory*. Isaac's Decision Support AI says "recommendations" but doesn't guarantee explainability.
- **EU AI Act**: High-risk AI requires "transparency obligations" (Art. 13) and documentation of design choices
- **NIST AI RMF**: "Transparency" is a core trustworthiness characteristic (MAP function)
- **ISO 42001**: §5.3.5 requires "understandable information about AI system decision-making"

Why missing: Isaac conflates "explanation" with governance (human review). Explainability is a *capability*, not a control.

What's needed:

- Standalone capability: **Explainability & Interpretability**

- Scope: Feature importance, decision rules, confidence intervals, reasoning chains
- Methods: LIME, SHAP, attention mechanisms, rule extraction
- Governance anchor: TBS ADM Directive §3.2 (citizen right to explanation)
- Example: "Regulatory triage AI explains: 'High priority (89%) because: contains 'adverse event' (weight 0.42) + references rare condition (weight 0.31)'"

Artifact:

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Explainability & Interpretability Capability
├ Feature Attribution (LIME, SHAP, attention)
├ Decision Rule Extraction (if-then rules, decision trees)
├ Confidence & Uncertainty Quantification
├ Counterfactual Explanations (what would change the decision)
└ Transparency Documentation (for audit, citizen disclosure)

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Gap 1.3: Bias & Fairness Testing & Detection

Signaled by:

- **NIST AI RMF:** "Fairness" is explicitly mapped in MEASURE and MANAGE functions
- **EU AI Act:** High-risk AI requires "bias testing and mitigation" (Art. 10)
- **ISO 42001:** §5.3.3 requires "identification and mitigation of AI-specific risks including bias"
- **WHO AI Guidance:** "Equity" is a core principle for health AI
- **HC/PHAC EA:** Regulatory and surveillance AI must not perpetuate health inequities

Why missing: Bias detection is a *prerequisite* before deployment, not covered by Isaac's 8 capabilities.

What's needed:

- Standalone capability: **Bias & Fairness Testing**
 - Scope: Demographic parity testing, disparate impact analysis, fairness metrics (equalized odds, calibration), debiasing
 - Governance: Pre-deployment gate, continuous monitoring
 - Example: "Classification AI tested for accuracy disparity across age groups; remediated via balanced training data (now 91% accuracy for all age bands)"

Artifact:

Bias & Fairness Testing Capability

- └ Demographic Parity Analysis (equal prediction rates across groups)
 - └ Disparate Impact Testing (4/5 rule compliance)
 - └ Fairness Metrics (equalized odds, calibration, fairness-accuracy tradeoff)
 - └ Bias Mitigation (re-weighting, synthetic data, debiasing algorithms)
 - └ Continuous Monitoring (detect fairness drift in production)
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Gap 1.4: Enterprise Integration & API Governance

Signaled by:

- **PSPC:** "AI services" sold to GC require clear integration contracts, API specifications, SLA guarantees
- **HC/PHAC EA:** "Platform reuse" means AI capabilities must plug into HAIL with standardized interfaces
- **ISO 42001:** §5.4.3 requires "documented interfaces" and "integration controls"

Why missing: Isaac describes AI *models* but not the *integration layer* that makes them reusable.

What's needed:

- Standalone capability: **Enterprise AI Integration**
 - Scope: API contracts, request/response schemas, latency/throughput SLAs, error handling, rate limiting, observability hooks
 - Governance: Service-level agreements (SLAs), change management (breaking API changes), security (OAuth, encryption)
 - Example: "Generative AI service: REST API, 2s latency p95, 99.9% uptime SLA, rate limit 100 req/min per client, integrates with HAIL audit logger"

Artifact:

Enterprise AI Integration Capability

- └ API Contract & Specification (OpenAPI/Swagger)
- └ Service Level Agreements (latency, throughput, uptime)
- └ Error Handling & Fallback (graceful degradation)
- └ Observability & Logging (integrates with audit trail)
- └ Security (authentication, encryption, rate limiting)

└ Change Management (backward compatibility, deprecation path)

Gap 1.5: Real-time & Streaming AI

Signaled by:

- **HC/PHAC EA:** "Surveillance" cluster requires *real-time* anomaly detection, not batch predictions
- **GC ADM Directive:** "Timely" review of automated decisions (implies near-real-time trigger + approval)
- **WHO AI Guidance:** Public health surveillance must detect signals within hours, not days

Why missing: Isaac's Predictive AI focuses on forecasting (batch). Real-time AI is a distinct capability.

What's needed:

- Standalone capability: **Real-time & Streaming AI**
 - Scope: Event-driven decision-making, streaming ML, drift detection, online learning, sub-100ms inference
 - Governance: Low-latency audit trails, escalation triggers, circuit breakers
 - Example: "Surveillance AI: processes health event stream (1000 events/sec), detects anomalies in <5sec, escalates to epidemiologist for human review"

Artifact:

Real-time & Streaming AI Capability

- └ Event Stream Processing (Kafka, Azure Event Hubs)
 - └ Online Learning (incremental model updates)
 - └ Drift Detection (real-time performance monitoring)
 - └ Sub-second Inference (edge deployment, model quantization)
 - └ Escalation Workflows (trigger human review on high-confidence anomalies)
 - └ Low-latency Audit Trail (audit log write < 100ms)
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Gap 1.6: Data Privacy & Isolation Controls (AI-specific)

Signaled by:

- **SSC:** "Secure enterprise platforms" host multi-tenant AI. AI-specific privacy controls

needed (federated learning, differential privacy, secure enclaves)

- **HC/PHAC EA:** Protected B health data requires isolation from other tenants
- **EU AI Act:** Article 10 (high-risk AI) requires data minimization and access controls
- **NIST AI RMF:** "Privacy" is a core trustworthiness characteristic

Why missing: Isaac doesn't address how AI handles sensitive data in shared infrastructure.

What's needed:

- Standalone capability: **Data Privacy & Isolation (AI-specific)**
 - Scope: Federated learning (train without centralizing data), differential privacy (add noise to protect individuals), secure enclaves, data minimization, synthetic data generation
 - Governance: Data governance for ML (who can access training data, lineage tracking)
 - Example: "Predictive AI trained via federated learning: each health region trains locally, model updates aggregated without sharing raw data"

Artifact:

Data Privacy & Isolation Capability

- └ Federated Learning (distributed training without data centralization)
- └ Differential Privacy (add noise to protect individual privacy)
- └ Synthetic Data Generation (replace real data with privacy-preserving surrogate)
- └ Secure Enclaves (hardware-based data isolation for training)
- └ Data Minimization (use only necessary features)
- └ Access Controls (role-based access to training data and models)

Gap 1.7: Adversarial Robustness & Security Testing

Signaled by:

- **NIST AI RMF:** "Robustness" and "security" are core trustworthiness characteristics
- **ISO 42001:** §5.3.4 requires testing for "AI-specific security risks" (data poisoning, model inversion, adversarial examples)
- **SSC:** Cloud-hosted AI must withstand adversarial attacks (prompt injection, model extraction)

Why missing: Isaac assumes models are secure. Adversarial robustness is a capability, not a given.

What's needed:

- Standalone capability: **Adversarial Robustness Testing**
 - Scope: Adversarial example generation, prompt injection testing, model extraction defense, data poisoning resistance
 - Governance: Pre-deployment security testing gate
 - Example: "Generative AI tested for prompt injection (100+ attack patterns), model extraction (API queries logged, rate limited), adversarial suffixes (FGSM, PGD attacks)"

Artifact:

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Adversarial Robustness & Security Testing Capability
├─ Adversarial Example Generation (FGSM, PGD, C&W attacks)
├─ Prompt Injection Testing (jailbreak attempts, instruction overrides)
├─ Model Extraction Defense (rate limiting, query obfuscation, output perturbation)
├─ Data Poisoning Resistance (training data validation, outlier detection)
├─ Backdoor Detection (neural network trojan testing)
└─ Red Team Exercises (simulated adversarial use)
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TIER 2: RE-FRAMING NEEDS — Isaac Capabilities Need Explicit Sub-Capabilities

Gap 2.1: Decision Support AI → Add "Auditability & Logging" as explicit sub-capability

Signaled by:

- **TBS ADM Directive** (§3.1): Every automated decision must have "an explanation", "a method to appeal", and "records of the decision"
- Current Isaac framing: "Recommendations, prioritization" — implies human review but doesn't guarantee auditability

What's needed:

Decision Support AI

- | Recommendations & Prioritization (current)
- | Explainability (see Gap 1.2 above)
- | ****NEW: Auditability & Logging****
 - | | Decision audit trail (who made decision, when, based on what input)
 - | | Appeal & Override Logging (human overrode recommendation, reason)
 - | | Regulatory Compliance Logging (evidence for TBS audit, ATIP requests)
- | Human-in-Loop Controls (approval workflows, escalation triggers)

Governance profile update:

Current: "Human-in-loop governance"

Revised: "Human-in-loop governance + audit trail + appeal records + regulatory di

Gap 2.2: Orchestration AI → Add "Change Management & Rollback" as explicit sub-capability

Signaled by:

- **PSPC:** Vendors selling agentic AI services need clear rollback procedures and change testing protocols
- **HC/PHAC EA:** Workflow automation touching regulated processes (e.g., PMRA submissions) requires safe deployment
- **ISO 42001:** §5.4.7 requires "change control procedures" for AI updates

What's needed:

Orchestration AI

- | Agentic Workflows & Task Automation (current)
- | Human Override & Monitoring (current)
- | ****NEW: Change Management & Rollback****
 - | | Staged rollout (canary, shadow mode, A/B testing)
 - | | Automated rollback triggers (error rate threshold, latency spike)
 - | | Testing in pre-production (workflow regression testing)
 - | | Communication Plan (notify users of changes, deprecation warnings)
- | Safety Guarantees (circuit breakers, fail-safe states)

Gap 2.3: Perception AI → Split into "Structured" and "Unstructured" variants

Signaled by:

- **HC/PHAC EA:** OCR (document processing) and vision (lab images) have different governance profiles
- Current Isaac: “OCR, vision, speech processing” lumped together — may mask different risk/control needs

What’s needed:

Perception AI (Structured)

- └ Document Digitization (OCR, form field extraction)
- └ Quality Assurance (low-confidence detection, manual review routing)
- └ Governance: High accuracy required (e.g., 99%+ for regulatory documents)

Perception AI (Unstructured)

- └ Image Analysis (medical imaging, lab samples, scene understanding)
- └ Confidence Thresholding & Escalation (route low-confidence to specialist)
- └ Governance: Bias testing required (imaging AI often trained on non-diverse data)

TIER 3: HC/PHAC-Specific Domain Capabilities (Not GC-wide, but HC signals them)

These are signaled by **HC/PHAC EA** and **PMRA program**, but not by TBS/PSPC/SSC. They may belong in a separate “HC/PHAC Domain Specializations” section, not in the core 8.

Gap 3.1: Scientific Computing AI

Signaled by:

- **HC/PHAC EA briefing:** “RAG, predictive analytics, scientific AI”
- **PMRA Agent Briefing:** Literature review automation, regulatory submission intelligence, scientific paper triage
- **WHO AI Guidance:** “Clinical AI” specific to health domain

Scope: Lab collaboration workflows, literature synthesis, hypothesis generation, experimental design support

Governance profile: Domain-specific validation (peer review, reproducibility), IP disclosure (patents, publications)

Is this a distinct capability? Maybe not — it’s likely **Knowledge AI (RAG) + Generative AI (synthesis)** applied to scientific domain. But HC signals it as strategically important.

Gap 3.2: Epidemiological Signal Detection AI

Signaled by:

- **HC/PHAC EA briefing:** "Scale and Strengthen Surveillance" focus area
- **Surveillance cluster** definition: Real-time health signal detection
- **WHO AI Guidance:** Early warning systems for outbreaks

Scope: Real-time anomaly detection across health data sources, outbreak prediction, signal quality assessment

Is this a distinct capability? Arguably **Predictive AI (forecasting, anomaly detection)** + **Real-time AI (streaming)** + **Data Privacy (federated learning across provinces)**. But HC signals it as mission-critical.

Gap 3.3: Regulatory Intelligence AI

Signaled by:

- **PMRA program:** "Regulatory submission intelligence" — extract structured data from scientific literature, regulatory filings, manufacturing specs
- **HC AI Strategy:** "Enhance Regulatory Processes and Services" focus area
- Not GC-wide; specific to HC/PHAC regulatory mandate

Scope: Literature triage, data extraction from PDFs, regulatory requirement matching, submission risk assessment

Is this a distinct capability? Likely **Classification AI (automated tagging, triage)** + **Perception AI (document digitization)** + **Knowledge AI (semantic search across regulations)**. But HC signals it as a procurement priority.

SUMMARY TABLE: Isaac vs. GC Signals

Gap	Type	Signaled By	Isaac Status	Recommendation
1.1	Model Management &	PSPC, HC/PHAC EA,	Missing	Add as standalone

	Registry	ISO 42001		capability
1.2	Explainability & Interpretability	TBS ADM, EU AI Act, NIST, ISO 42001	Missing	Add as standalone capability
1.3	Bias & Fairness Testing	NIST, EU AI Act, ISO 42001, WHO	Missing	Add as standalone capability
1.4	Enterprise Integration & API	PSPC, HC/PHAC EA, ISO 42001	Missing	Add as standalone capability
1.5	Real-time & Streaming AI	HC/PHAC EA (Surveillance), ADM Directive	Missing	Add as standalone capability
1.6	Data Privacy & Isolation (AI-specific)	SSC, HC/PHAC EA, EU AI Act, NIST	Missing	Add as standalone capability
1.7	Adversarial Robustness Testing	NIST, ISO 42001, SSC	Missing	Add as standalone capability
2.1	Decision Support AI re-frame	TBS ADM Directive	Partial	Add "Auditability & Logging" sub-capability
2.2	Orchestration AI re-frame	PSPC, HC/PHAC EA, ISO 42001	Partial	Add "Change Management & Rollback" sub-capability
2.3	Perception AI split	HC/PHAC EA	Partial	Split into Structured & Unstructured variants
3.1	Scientific Computing AI	HC/PHAC EA, PMRA	HC-specific	Consider "HC/PHAC Domain Specializations" section
3.2	Epidemiological Signal Detection AI	HC/PHAC EA, Surveillance cluster	HC-specific	Consider "HC/PHAC Domain Specializations" section
3.3	Regulatory Intelligence AI	PMRA, HC AI Strategy	HC-specific	Consider "HC/PHAC Domain Specializations" section

Revised Isaac Framework (15 Capabilities)

Core GC AI Capabilities (8 → 15):

1. Generative AI ✓
2. Knowledge AI ✓
3. Predictive AI ✓
4. Classification AI ✓
5. Conversational AI ✓
6. Perception AI ✓ (split into Structured & Unstructured)
7. Orchestration AI ✓ (+ Change Management sub-cap)
8. Decision Support AI ✓ (+ Auditability sub-cap)
9. **NEW: AI Model Management & Registry**
10. **NEW: Explainability & Interpretability**
11. **NEW: Bias & Fairness Testing**
12. **NEW: Enterprise AI Integration**
13. **NEW: Real-time & Streaming AI**
14. **NEW: Data Privacy & Isolation**
15. **NEW: Adversarial Robustness Testing**

HC/PHAC Domain Specializations (optional, not in core):

- Scientific Computing AI
- Epidemiological Signal Detection AI
- Regulatory Intelligence AI

Materiality Assessment for ARB Submission

High Risk (ARB will flag):

- Gaps 1.2, 1.3 (Explainability, Bias) — TBS and EU Act are non-negotiable
- Gap 1.1 (Model Management) — ISO 42001 compliance requires this

Medium Risk:

- Gap 1.4 (Integration) — PSPC will challenge vendor selection without this
- Gap 2.1 (Auditability) — TBS ADM Directive compliance

Lower Risk but operationally important:

- Gaps 1.5, 1.6, 1.7 — HC/PHAC EA signals but not explicitly required by TBS/PSPC policy yet
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Next Steps

1. **Confirm scope:** Are we proposing 8 capabilities (TBS/PSPC-focused) or 15 (TBS/PSPC + ISO 42001 + HC/PHAC EA)?
2. **Add to spreadsheet:** New sheets for Gaps 1.1–1.7 with governance profiles, policy anchors, HAIL integration points
3. **Validate against ARB checklist:** Does revised framework close TBS ADM Directive, EU AI Act, ISO 42001 compliance doors?
4. **Decide on HC specializations:** Do Gaps 3.1–3.3 go in the core framework or a separate “HC/PHAC Capability Extensions” section?